

ADA-Marker Answer Sheet

Anonymized local demo

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PROBLEM

Q2

Answer

5. 對回文, 從 median index 切左=右 (反讀), we can delete but not insert element. 因此切一半 (奇偶沒差, median 一定會一樣), Set $T=0, J=\lfloor n/2 \rfloor$ if $T=J$

$T++, J--$ / if $T \neq J$ $\left[\begin{array}{l} T+1=J \text{ then delete } i \\ T=J-1 \text{ then delete } j \end{array} \right]$ 使得 $T=J$ for all 左半 & 右半
(T from $0 \sim \lfloor \frac{n}{2} \rfloor$)

$\left[\begin{array}{c} \text{左} \quad \text{右} \end{array} \right] \Rightarrow \begin{array}{c} i=0 \quad 1 \quad 2 \quad 3 \quad 4 \\ \boxed{} \boxed{} \boxed{} \boxed{} \boxed{} \end{array} \quad \begin{array}{c} 0 \quad 1 \quad 2 \quad 3 \quad 4 \\ \boxed{} \boxed{} \boxed{} \boxed{} \boxed{} \end{array} = j$
(J from $\lfloor \frac{n}{2} \rfloor$ to 0)

$\Rightarrow O(\frac{n}{2}) \times O(\frac{n}{2}) = \frac{1}{4} O(n^2) = O(n^2) \Rightarrow$ cannot ensure longest X

★ Use string alignment algo 加上 "-" 找出原始 S 跟 $reverse(S)$ 重複最長的區間
其中 if S, S^+ match +2

用 $S = 0 \ 0 \ 0 \ 0 \ 0 \dots$ not match -2, "-" -1, ~~1~~

DP use S_0 $0 \rightarrow 0$ 計算每個 node 當下最優解可得兩串重複最長

$0 \rightarrow 0$

0

0

0

0

0

0

0

0

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0

0

0

把 "-" 外的 delete 即得 longest 回文

work in $O(n^2)$ time (動態正反的 string seq)
 $O(n) \times O(n)$